Luis Lopez Rondon

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Education

University of South Florida, Judy Genshaft Honors College — **Tampa, FL**
Bachelor of Science in Electrical Engineering — ***GPA: 4.00*** | *May 2029*

Honors: National Merit Finalist, Sunshine State Scholar

Technical Skills

Hardware	STM32 (CubeIDE, CubeMX), ESP32, Teensy, soldering (SMD, THT)
Instrumentation	Oscilloscope, DMM, logic analyzer, bench power supply
EDA & Simulation	Altium Designer, KiCad, LTspice
Programming	C/C++, MATLAB
Spoken Languages	English, Spanish

Technical Experience

IEEE at USF EXO — **Tampa, FL**
Electrical Systems Lead, Lower-Limb Exoskeleton — *Aug 2026 – Present*

- Leading electrical development of a **unilateral exoskeleton** with two 48V CubeMars AK80-9 actuators for hip and knee assistance and a passive ankle mechanism.
- Architecting power for the embedded control stack (**Jetson Orin Nano**, **Teensy 4.1**), including 48V-to-logic DC-DC rails and a relay-based **hardware e-stop** that cuts power from the motor bus.
- Modeled motor and battery current in **MATLAB** from OpenSim-derived gait data (**19.49A** motor peak; ∼**10A** battery peak vs. a >50A initial estimate) to size the 48V Li-ion pack.
- Developing gait-control and verification procedures using **joint encoders**, **IMUs**, and **heel/toe force sensors** to time assistance across gait-phase transitions.

Projects

PID-Stabilized Parkinson's Spoon | *IEEE at USF EXO* — *Aug 2026 – Present*

- Designed the PCB for a PID-controlled Parkinson's spoon around an **STM32G431** and **LSM6DSV16X IMU**.
- Built the power path from a single 1S LiPo with **TP4056** USB-C charging, DW01A/FS8205 protection, and a custom 5V boost converter for the stabilizing servo.
- Taught the design in an **Altium workshop** covering part selection, datasheet reading, and layout, with ∼10 EXO members each completing their first board (a 6-axis IMU breakout).

Audio Spectrum Analyzer — *Aug 2026 – Present*

- Built a real-time analyzer in **C++** on a **Teensy 4.0 + SGTL5000 codec** (I2S audio, I2C control) with USB/line-in input, a **1024-point FFT** in 16 log-spaced bands, and an ILI9341 SPI TFT.
- Validated the analyzer's frequency response by comparing its band output against an **oscilloscope FFT** of the same input signal.
- Designing a custom **STM32F411** PCB integrating the SGTL5000 codec and the TFT display, with an AC-coupled line-in, filtered analog supply, and **mixed-signal layout.**

Mime: Mimetic Robotic Hand | *IEEE at USF EXO* — *May 2026 – Aug 2026*

- Designed an **ESP32-S3** + **PCA9685** PCB in Altium for a hand driven by a flex glove and OpenCV.
- Tested glove and OpenCV control on a breadboard prototype before committing to a custom PCB.